Discrete Structure (CST-002) - Unit 3

Unit 3 - Counting & Recurrence Relations - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: counting techniques and recurrence relations - how to count without listing, and how to solve sequences defined by their previous terms.

Key ideas: Product rule (choices multiply for "and"), **sum rule** (choices add for exclusive "or"). **Permutation** P(n,r) = n!/(n-r)! when order matters; **combination** C(n,r) = n!/(r!(n-r)!) when it does not. **Pigeonhole:** n+1 pigeons in n holes force one hole with at least two - great for short proof questions. **Inclusion-exclusion:** $|A \cup B| = |A| + |B| - |A \cap B|$. For **recurrence relations** like $a_n = 5a_{n-1} - 6a_{n-2}$: write the characteristic equation, find roots r_1, r_2; distinct roots give $a_n = A \cdot r_1^n + B \cdot r_2^n$, repeated root r gives $a_n = (A + Bn)r^n$; fix A, B from the initial conditions.

Exam pointers: solving a linear homogeneous recurrence with given initial conditions is the single most repeated numerical - practise 4-5 until the method is automatic. Know the difference between "arrangement" (permutation) and "selection" (combination) word problems.